

Lab 01: The Rental Refund Crisis

Query Query History

1 `SELECT rental_id, customer_id FROM rental LIMIT 1;`
2

Data Output Messages Notifications

≡+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

Showing rows: 1 to 1

✎

🖨️

 Page No: 1 of 1

⏪

⏴

⏵

⏩

	rental_id [PK] integer ✎	customer_id smallint ✎
1	2	459

1 `SELECT rental_id, customer_id FROM rental LIMIT 1;`
2
3 `SELECT rental_id, return_date FROM rental WHERE rental_id = 2;`
4

Data Output Messages Notifications

≡+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

Showing rows: 1 to 1

✎

🖨️

 Page No: 1 of 1

	rental_id [PK] integer ✎	return_date timestamp without time zone ✎
1	2	2005-05-28 19:40:33

Exercise 1: Testing the "All or Nothing"

```
5 BEGIN;
6 UPDATE rental SET return_date = NOW() WHERE rental_id = 2;
7
8 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
9
10 VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
11
12 COMMIT;
13
14 ROLLBACK;
15
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

```
14 ROLLBACK;
15
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 46 msec.

```
13
14 ROLLBACK;
15
16 SELECT return_date FROM rental WHERE rental_id = 2;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	return_date timestamp without time zone
1	2005-05-28 19:40:33

Savepoints: Partial Rollbacks

```

18 BEGIN;
19 INSERT INTO staff (first_name, last_name, address_id, email, store_id, active, username)
20 VALUES ('Trang', 'Student', 1, 'trang@student.com', 1, true, 'trang_student');

```

Data Output Messages Notifications

INSERT 0 1

Query returned successfully in 47 msec.

```

22 SAVEPOINT staff_created;

```

Data Output Messages Notifications

SAVEPOINT

Query returned successfully in 38 msec.

```

22 SAVEPOINT staff_created;

```

```

23
24 INSERT INTO special_offers (customer_id, code) VALUES (9999, 'FREE')

```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

```

24 INSERT INTO special_offers (customer_id, code) VALUES (9999, 'FREE')

```

```

25
26 ROLLBACK TO staff_created;

```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 35 msec.

Exercise 2: The "Flexible" Staff Task

```

30 ALTER TABLE staff
31 ADD CONSTRAINT staff_store_fk
32 FOREIGN KEY (store_id)
33 REFERENCES store(store_id);
34
35 BEGIN;
36 INSERT INTO staff (first_name, last_name, address_id, email, store_id, active, username)
37
38 VALUES ('AI', 'Student', 1, 'ai@student.com', 1, true, 'ai_student');
39
40 SAVEPOINT staff_created;

```

```
30 BEGIN;  
31 INSERT INTO staff (first_name, last_name, address_id, email, store_id, active, username)  
32  
33 VALUES ('AI', 'Student', 1, 'ai@student.com', 1, true, 'ai_student');
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
INSERT 0 1

Query returned successfully in 36 msec.

```
35 SAVEPOINT staff_created;  
36
```

Data Output Messages Notifications

SAVEPOINT

Query returned successfully in 28 msec.

```
42 UPDATE staff SET store_id = 9999 WHERE username = 'ai_student';  
43  
44 ROLLBACK TO staff_created;  
45  
46 UPDATE staff SET store_id = 1 WHERE username = 'ai_student';  
47
```

Data Output Messages Notifications

ERROR: insert or update on table "staff" violates foreign key constraint "staff_store_fk"
Key (store_id)=(9999) is not present in table "store".

SQL state: 23503

Detail: Key (store_id)=(9999) is not present in table "store".

```
44 ROLLBACK TO staff_created;  
45  
46 UPDATE staff SET store_id = 1 WHERE username = 'ai_student';  
47  
48 COMMIT;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 38 msec.

```
46 UPDATE staff SET store_id = 1 WHERE username = 'ai_student';
47
48 COMMIT;
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 40 msec.

```
48 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 29 msec.